Data and Algorithm Analysis

Chapter 3 — Asymptotic Notation

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Outline

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Overview

- Describe behavior of functions in the limit.
- Describe growth of functions.
- Abstract away low-order terms and constant factors.
- How to determine the running times of algorithms .

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Comparing the Growth Rate of Functions

Start with any two functions $f, g : \mathbb{N} \to \mathbb{Z}$ (or $\mathbb{N} \to \mathbb{R}$).

Want to relate the growth rates of $f(n)$ and $g(n)$ asymptotically (as $n \to \infty$) within constant factors.

Notation	Approximate Meaning	Relation
O	"less than or same"	$\leq$
Ω	"greater than or same"	$\geq$
Θ	"same as"	$=$
o	"less than"	$>$
ω	"greather than"	$<$

Advantages of Asymptotic Notation

1. Avoid the tedious computation of $T(n)$ in detail.
2. Allows for comparison of two functions $f(n)$ and $g(n)$:
   - Ignoring constant factors.
   - Only as $n \to \infty$.
3. Makes our lives easier!

O Notation

$$O(g(n)) = \{f(n) \mid \text{there exist positive constants } c, n_0 \text{ such that } 0 \leq f(n) \leq cg(n) \text{ whenever } n \geq n_0\}$$

Write

$$f(n) \in O(g(n))$$

or

$$f(n) = O(g(n)).$$

Ω Notation

$$\Omega(g(n)) = \{f(n) \mid \text{there exist positive constants } c, n_0 \text{ such that } 0 \leq cg(n) \leq f(n) \text{ whenever } n \geq n_0\}$$

Write

$$f(n) \in \Omega(g(n))$$

or

$$f(n) = \Omega(g(n)).$$

Θ Notation

$$\Theta(g(n)) = \{f(n) \mid \text{there exist positive constants } c_1, c_2, n_0 \text{ such that } 0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n) \text{ whenever } n \geq n_0\}$$

Write

$$f(n) \in \Theta(g(n))$$

or

$$f(n) = \Theta(g(n)).$$

Observations

- $f(n) = \Theta(g(n))$ if and only if $g(n) = \Theta(f(n))$.
- $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$ is equivalent to $f(n) = \Theta(g(n))$.
- Not all f and g are related in asymptotic growth rate, such as

$$\begin{aligned} f(n) &= n^7 \\ g(n) &= \begin{cases} n^6 & \text{if } n \text{ is even;} \\ n^8 & \text{if } n \text{ is odd.} \end{cases} \end{aligned}$$

Examples — Asymptotic Notation

$$
\begin{aligned}
3n^2 - 7n + 14 &= \Theta(n^2) \\
n^4 \lg n + \frac{3n^2}{\lg n} - 5 &= \Omega(n^3) \\
n^9 - 18n^5 &= O(n^{12}) \\
7 \cdot 2^{n+1} &= \Theta(2^n)
\end{aligned}
$$

Asymptotic Analysis by Limits

If $\lim_{n\to\infty} \frac{f(n)}{g(n)} = C$ is defined, then we have

- if $C = 0$, then $f(n) = O(g(n))$ but $f(n) \neq \Theta(g(n))$;
- if $C = \infty$, then $f(n) = \Omega(g(n))$ but $f(n) \neq \Theta(g(n))$; and
- if C is a non-zero constant, then $f(n) = \Theta(g(n))$.

Since the limit is not always defined, this does not always work.

Exercise: What is $\lim_{n\to\infty} \frac{2^n}{n!}$?

Hierarchies of Growth Rates

There are several common patterns for the asymptotics of time complexities that arise frequently.

$$1 \quad \lg n \quad \lg^2 n \quad \cdots \quad n \quad n \lg n \quad n \lg^2 n \quad \cdots$$

$$n^2 \quad n^3 \quad \cdots \quad 2^n \quad 3^n \quad \cdots \quad n! \quad \cdots$$

Each successive function grows strictly faster than its predecessors.

Asymptotic Notation vs. Running Time

Asymptotic notation describes functions, which in turn describe running times.

Must be careful to specify **which** running time.

Use asymptotic notation **correctly** when characterizing a running time.

Asymptotic Notation vs. Running Time

For insertion sort

- *Worst-case* running time is $O(n^2)$, $\Omega(n^2)$, or $\underline{\Theta(n^2)}$
- *Best-case* running time is $O(n)$, $\Omega(n)$, or $\underline{\Theta(n)}$

Cannot say the running time for insertion sort is $\Theta(n^2)$ (with "worst-case" omitted).
Can say the running time for insertion sort is $O(n^2)$, or $\Omega(n)$.

For merge sort, its running time is $\Theta(n \log n)$ in *all* cases, so it's OK to omit which case.

Asymptotic Notation vs. Running Time

Common error: Conflating O-notation with Θ-notation by using O-notation to indicate an asymptotically **tight** bound.

Saying an $O(n \log n)$-time algorithm runs faster than an $O(n^2)$-time algorithm is not necessarily true.

- An $\Theta(n)$-time algorithm also runs in $O(n^2)$ time.

If it is really mean an asymptotically tight bound, then use Θ-notation.

Example: $3n^2 + 20n$

- $\Theta(n^2)$ is the most accurate. $O(n^3)$ correct but less precise.

Asymptotic Notation in Equations

When on RHS, $O(n^2)$ means some anonymous function in the set $O(n^2)$.

Example: $2n^2 + 3n + 1 = 2n^2 + \Theta(n)$

- $2n^2 + 3n + 1 = 2n^2 + f(n)$ for some $f(n) \in \Theta(n)$ (i.e., $f(n) = 3n + 1$).

When on LHS, no matter how the anonymous functions are chosen on LHS, there is a way to choose the anonymous functions on RHS to make the equation valid.

Example: $2n^2 + \Theta(n) = \Theta(n^2)$

- $\forall f(n) \in \Theta(n)$, there exists a function $g(n) \in \Theta(n^2)$ s.t. $2n^2 + f(n) = g(n)$.

Asymptotic Notation in Equations

Can also chain together:

$$\begin{aligned} 2n^2 + 3n + 1 &= 2n^2 + \Theta(n) \\ &= \Theta(n^2) \end{aligned}$$

First equation: There exists $f(n) \in \Theta(n)$ s.t. $2n^2 + 3n + 1 = 2n^2 + f(n)$.

Second equation: $\forall g(n) \in \Theta(n)$ (s.t. $f(n)$ holds in the first), there exists $h(n) \in \Theta(n^2)$ s.t. $2n^2 + g(n) = h(n)$.

o-notation

$$
\begin{aligned}
o(g(n)) \quad = \quad & \{f(n) : \text{for all constants } c > 0 \\
& \quad \text{there exists a constant } n_0 > 0 \text{ such that} \\
& \quad 0 \leq f(n) < cg(n) \\
& \quad \forall n \geq n_0\}.
\end{aligned}
$$

Easier view to use: $\lim_{n\to\infty} \frac{f(n)}{g(n)} = 0$

- $n^{1.9999} = o(n^2)$
- $n^2 \lg n = o(n^2)$
- $n^2 \neq o(n^2)$
- $n^2/1000 \neq o(n^2)$

ω-notation

$$\begin{aligned}\omega(g(n)) \quad = \quad \{f(n) : & \text{ for all constants } c > 0 \\ & \text{there exists a constant } n_0 > 0 \text{ such that} \\ & 0 \leq cg(n) < f(n) \\ & \forall n \geq n_0\}.\end{aligned}$$

Easier view to use: $\lim_{n\to\infty} \frac{f(n)}{g(n)} = \infty$

- $n^{2.0001} = \omega(n^2)$
- $n^2 \log n = \omega(n^2)$
- $n^2 \neq \omega(n^2)$

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Comparisons of Functions

Transitivity

- $f(n) = \Theta(g(n))$ and $g(n) = \Theta(h(n)) \Rightarrow f(n) = \Theta(h(n))$
- Same for O, Ω, o, ω

Reflexivity

- $f(n) = \Theta(f(n))$
- Same for O and Ω

Symmetry

- $f(n) = \Theta(g(n))$ iff $g(n) = \Theta(f(n))$

Transpose symmetry

- $f(n) = O(g(n))$ iff $g(n) = \Omega(f(n))$
- $f(n) = o(g(n))$ iff $g(n) = \omega(f(n))$

Comparisons of Functions

Comparisons:

- $f(n)$ is **asymptotically smaller** than $g(n)$ if $f(n) = o(g(n))$.
- $f(n)$ is **asymptotically larger** than $g(n)$ if $f(n) = \omega(g(n))$.

No trichotomy. Although intuitively, we can liken O to $\leq$, Ω to $\geq$, etc., we might not be able to compare functions.

Example: $n^{1+\sin n}$ vs. n

Logarithms

Notations:

$\lg n = \log_2 n$ (binary logarithm)
$\ln n = \log_e n$ (natural logarithm)
$\lg^k n = (\lg n)^k$ (exponentiation)
$\lg \lg n = \lg(\lg n)$ (composition)

In the expression $\log_b a$:

- Hold b constant $\Rightarrow$ the expression is strictly increasing as a increases.
- Hold a constant $\Rightarrow$ the expression is strictly increasing as b increases.

Logarithms

- $a = b^{\log_b a}$
- $\log_c(ab) = \log_c a + \log_c b$
- $\log_b a^n = n \log_b a$
- $\log_b a = \frac{\log_c a}{\log_c b}$
- $\log_b(1/a) = -\log_b a$
- $\log_b a = \frac{1}{\log_a b}$
- $a^{\log_b c} = c^{\log_b a}$